

How much signal-to-noise ratio a template bank costs, and how dense it has to be

J. Sfeir

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Abstract

We measure, in one controlled setting, the two quantities a template bank is designed against: the signal-to-noise ratio lost when a template does not match a signal, and the template density that loss implies. The setting is deliberately idealised — one detector, stationary Gaussian noise, an analytic power spectral density, non-spinning TaylorF2 templates in chirp-time coordinates — so that every number can be checked against an independent analytic calculation. Three results did not survive contact with the measurement. First, the hexagonal-to-square template ratio, conventionally compared against the constant-metric value 0.7698, is not a well-defined property of these banks: it runs from 0.4882 to 0.7796 in the same region depending only on how boundary templates are counted, because both regions are boundary-dominated at this covering radius. Second, the population-average volume loss is four to nine times smaller than the worst-case $1 - \text{MM}^3$ that is usually quoted. Third, and most consequentially, the effective number of independent trials grows as $N_{\text{templates}}^\alpha$ with $\alpha \approx 0.1568$, not as $N_{\text{templates}}$; refining a bank adds templates that overlap their neighbours by construction, so they add redundancy rather than trials. Under the naive trials count the effective detection volume falls with minimal match; under the calibrated one it rises. The sign of that trend, and with it the location of any optimum, is set by the trials model rather than by the bank.

1 What is idealised away, first

This is a controlled measurement, not a reproduction of a search. The idealisations are listed before the results because several of them bound what the results can mean.

- **One detector, Gaussian noise, analytic PSD.** `aLIGOZeroDetHighPower`, 20–1024 Hz. Real searches are limited by non-Gaussian transients, and handle them with signal-consistency tests and multi-detector coincidence. Nothing here models any of that, so no false-alarm rate reported below should be read as a search sensitivity.
- **Non-spinning TaylorF2, terminated at f_{ISCO} .** That cutoff is LAL’s default convention, not a physical statement, and it is responsible for most of the template-family error reported in §4.
- **The whole mass range lies above where post-Newtonian templates alone were trusted.** Advanced-LIGO PyCBC [5] used post-Newtonian templates only below $M = 4 M_\odot$. We report a main region $M \leq 35 M_\odot$ and an extended region $35 < M \leq 100 M_\odot$ separately, the latter being where the model visibly fails.
- **Mismatch is the quadratic form of a metric**, except where stated. The predictor was validated at $\mu = 0.03$ and fails at $\mu = 0.05$, so every curve below $\text{MM} = 0.97$ is outside the range in which the tool was checked.
- **Two parameters.** Chirp times (τ_0, τ_3) at $f_0 = 20 \text{ Hz}$. Production banks are four-dimensional and are not placed with a metric at all.

2 What is being measured

A template h recovers a fraction of the optimal signal-to-noise ratio equal to the match. In noise-free data that statement is exact, so the fractional SNR loss is exactly the mismatch $\mu = 1 - \langle \hat{h} | \hat{s} \rangle$. With noise the statistic $|z|$ is a Rice variable with $\langle |z|^2 \rangle = \rho_{\text{opt}}^2 \text{match}^2 + 2$; signal-free it satisfies $P(|z| > r) = e^{-r^2/2}$, verified on simulated noise.

A bank is then a set of templates dense enough that no signal in the region is further than a chosen mismatch from the nearest one. The chosen value, the minimal match, is a design convention. Owen [1] introduced it as a quantity “chosen by the experimenter based upon what he or she considers to be an acceptable loss of ideal event rate” and set the fiducial $\text{MM} = 0.9700$ to correspond to roughly 90 % of the ideal rate, since for sources uniform in Euclidean volume the rate goes as the cube of the SNR and $1 - \text{MM}^3 = 0.08733$. The convention has survived: the O4 GstLAL bank [9] still sets it at 97 %. But it is now applied as a *percentile*, not a worst case — that bank reports fitting factors above 97 % for 90 % of injections — and §5 shows why the distinction matters.

3 Method

Templates are placed with a local-metric lattice in the manner of Cokelaer [3]: each cell steps with the metric at its own position, hexagonally or on a square lattice aligned with the local eigenvectors at Owen’s spacing [1]. Two rules differ from the pre-registered specification, each for a measured reason.

Cells landing on the non-physical side of $\eta = 1/4$ are pushed back onto the line *after* reproduction, as Cokelaer [3] prescribes, not during it. Projecting during growth collapses a two-dimensional neighbourhood onto a curve and nothing re-spaces it: in an earlier bank the proper distance between neighbours along that line reached 0.680 against a covering radius of 0.173.

Fertility — whether a cell outside the region may still reproduce — is judged with the metric at the nearest *region boundary point*, not at the cell. Where TaylorF2 loses in-band support the metric at the cell degenerates, and under the original rule a template at $(215.95, 0.0695) M_{\odot}$, whose real match against anything in the region is below 0.1, scored as more fertile than a genuine neighbour.

A boundary strip within one covering radius of every region edge is then repaired greedily. This step is ours, not Cokelaer’s, and §5 reports what it costs, because it is large enough to change the conclusions drawn from template counts.

4 The loss

Against IMRPhenomD signals the TaylorF2 family loses 0.51486 of the detection volume in the extended region. Of that, 0.50779 is the continuous family alone, so the bank’s discretisation adds seven tenths of a percentage point. *Above $M = 35 M_{\odot}$ the waveform model, not the bank, is what costs the volume*, and no amount of refinement recovers it. In the main region the same decomposition gives 0.19320 total against 0.17580 from the family. Both legs are computed from the same injections and the same seed; an earlier version assembled the two numbers from different runs and overstated the gap. This is the clearest practical result in the paper and it sets the context for everything that follows: bank density is worth arguing about only where the signal model is adequate.

	main hex	main square	ext. hex	ext. square
templates	1772	2273	336	488
constant-metric ideal	458	595	95	124
fraction at or above MM	1.00000	1.00000	0.99933	1.00000
population volume loss	0.02088	0.01534	0.01285	0.00945

Table 1: The four banks at MM = 0.97. The covering fractions are measured with uniform injections; §5 explains why they are not the whole story.

5 The bank

Uniform sampling flatters a bank. Under 3000 uniform injections three of the four banks contain no injection below the minimal match at all (Figure 5). That test has very little power: it reaches 50 % detection probability only at an uncovered fraction of 2.3×10^{-4} , and the two holes it does find in the extended hexagonal bank carry an exact Poisson 95 % interval of $[8.1 \times 10^{-5}, 2.4 \times 10^{-3}]$. A clean covering fraction from uniform injections is close to no evidence at all. An adaptive search finds otherwise: the worst mismatch it reaches is 0.0322 in the main hexagonal bank and 0.0362 in the extended one, both above the 0.03 target and both reproduced independently. Both square banks survive the same search. Figure 2 makes the point directly: no cell of a 90×90 grid exceeds the target, and a point found by optimisation does.

The ratio is not a property of the lattices. The hexagonal-to-square count ratio is conventionally read against 0.7698, the constant-metric geometry, and against the 0.717–0.719 Cokelaer measures once the metric varies. We obtain 0.7796 in the main region and 0.6885 in the extended one — but counting only the lattice, without the boundary treatment, gives 0.4882 and 0.4225 instead. The boundary repair is 50.7 % of the main hexagonal bank and 17.0 % of the extended square one, and the ratio inherits that asymmetry. The reason is geometric: 0.79900 of main-region injections and 0.93833 of extended-region ones lie within one covering radius of a boundary. The regions are ribbons: the main one has proper area 35.7174 against a proper perimeter of 255.7431, where a disc of the same area would have 21.2, giving an effective width of 1.6127 covering radii and 2.0884 for the extended region. There is almost no interior in which an asymptotic, boundary-free ratio could be measured, and 0.7698 is an asymptotic, boundary-free number. At the $(5, 5) M_{\odot}$ corner the two bounding edges leave in the same chirp-time direction: differentiating the two mass rays gives positively parallel tangents, so the limiting angle is exactly zero. This is an analytic result, not a measurement, which is why it is stated in words and not cited from the registry; what is measured is the leading secant coefficient, 0.1032 degrees per solar mass of equal component increment, verified down to separations of 10^{-12} . It is a true cusp: both a finite-angle floor and numerical cancellation were explicitly refuted.

Against the constant-metric ideal the four banks carry 3.54 to 3.94 times as many templates. We decline to call that redundancy: for a finite region the bulk cell count is neither an optimum nor a lower bound, and explicit strip and disc counterexamples were constructed to show it. In a region this thin the number of templates is set by the boundary, not by the area.

6 The threshold, and how many trials there really are

At the pre-registered operating point — a false-alarm rate of $1/(100 \text{ yr})$ over one year — solving $\text{FAP} = 1 - (1 - p)^N$ exactly with $p = e^{-\rho^2/2}$ gives $\rho^* = 8.5996$ for the main hexagonal bank. The rare-tail Poisson form agrees to within double-precision rounding: the largest difference we can resolve is 3.4×10^{-16} relative, while the true mathematical difference at this operating

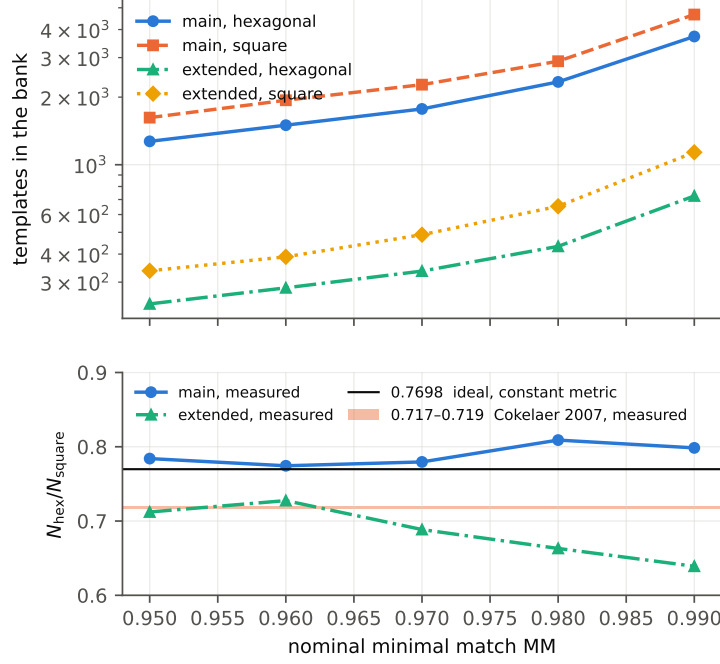


Figure 1: Bank size and the measured ratio against its two references. **How this could fail:** if lattice efficiency alone set the ratio, both regions would lie on one horizontal line near 0.77. They do not.

point is near 6×10^{-19} . The approximation is safe, but the measured number is floating-point noise rather than the error itself, and should not be quoted as if it were the latter.

Everything then depends on N . Counting every template and every sample as an independent trial is an upper bound, and provably so: for a finite bank-and-time grid in centred Gaussian noise the Gaussian correlation inequality [10] gives $\text{FAP}(\rho) \leq 1 - (1 - e^{-\rho^2/2})^N$. But it is a loose bound. Filtering simulated noise through each bank and taking the maximum of $|z|$ over templates and time gives $\nu_{\text{eff}} = 4.063 \times 10^5$ independent trials per second at $\text{MM} = 0.97$, between nine and thirty-one times below the naive count for the four banks at that minimal match. Within one region and lattice the gap *widens* as the bank is refined — 6.9 to 17.2 times across the MM grid for the main hexagonal family — but that is refinement, not size: pooled across all twenty banks the correlation between $\ln N_{\text{templates}}$ and the log gap is -0.45 , so with size alone the gap narrows.

The trial count is not constant across the tail it was fitted on: threshold-wise estimates taken at $|z|$ between 5.4 and 6.2 vary by factors of 1.8 to 5.2 across the twenty banks, against the single value used everywhere. The calibration is also made where the maxima fall, near $|z| = 5.3$ to 5.8, and applied at thresholds near 8.0 to 8.6 — an extrapolation of about two in $|z|$ that this work does not test. Because ρ^* depends on the trial count only logarithmically, a factor of two in ν_{eff} moves it by about 0.05; but it moves V_{eff} by a few per cent, against a calibrated trend of 0.2–1.8%, so this is a real limit on the strength of the conclusion in the next section and not a rounding remark.

That widening is the result. Fitting $N_{\text{eff}} \propto N_{\text{templates}}^\alpha$ within each family gives $\alpha = 0.1568$ (main hexagonal), 0.1500, 0.1516 and 0.2214. The weakest of the four sits 15.5 standard deviations below 1 and the strongest 23.4. The mechanism is not subtle: refining a bank adds templates that overlap their neighbours at the minimal match *by construction*, so they add redundancy, not independent trials.

The same twenty points fitted *together*, ignoring the split into regions and lattices, give $\alpha = 1.2922$ with a 95% interval of $[1.165, 1.424]$ — four standard deviations *above* 1, the opposite conclusion. We report it because a reader is entitled to it, and we do not use it: the

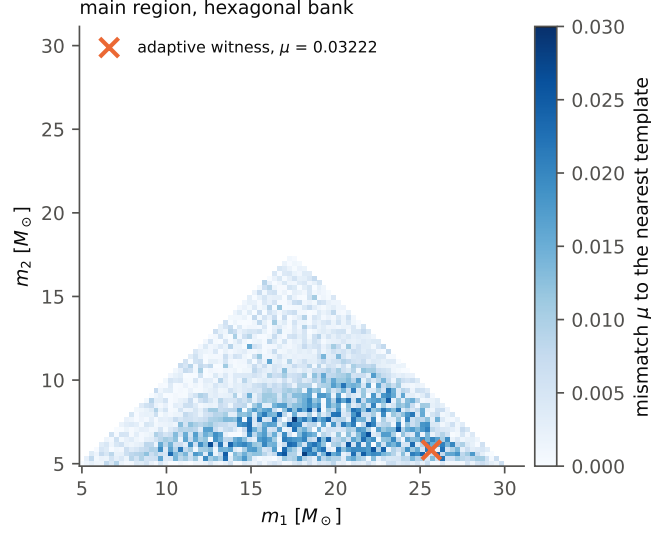


Figure 2: Mismatch to the nearest template over the mass plane, shown against both component masses because the metric depends on both. **How this could fail:** a covering bank would be uniformly below the target and none of this grid exceeds it — yet the marked point, found by an adaptive search, does. A figure built from grid sampling alone would have shown a covering bank.

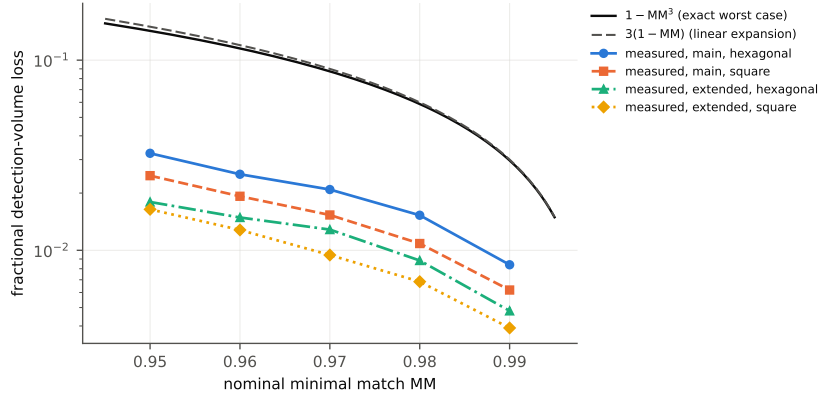


Figure 3: The three losses. **How this could fail:** if the worst case described what a bank costs, the measured points would lie on the black curve.

two regions differ in ν_{eff} by more than a factor ten at comparable template counts, so the pooled slope measures the step between them rather than how either one scales. The within-family fits are the ones that answer the question asked.

7 Volume, and the optimum that is not there

Three losses must be kept apart, and Figure 3 keeps them apart: the exact worst case $1 - \text{MM}^3 = 0.08733$, its linear expansion $3(1 - \text{MM})$, which is 3.06 % larger and is a different quantity, and the measured population average. The last is 0.02088 for the main hexagonal bank — *four to nine times smaller than the worst case*. Almost no injection lands at the worst point of its cell, so quoting $1 - \text{MM}^3$ as what a bank costs overstates it by most of an order of magnitude.

The effective volume $V_{\text{eff}} = \langle M^3 \rangle / \rho^{*3}$ then answers whether there is a best minimal match. Under the naive trials count it falls across $\text{MM} = 0.95\text{--}0.99$, by 1.87 % in the main hexagonal case; under the calibrated one it rises by 1.76 %. (The registry stores the naive change as a signed quantity, -1.87% .) The threshold is the reason: calibrated, ρ^* moves 0.24 % while the

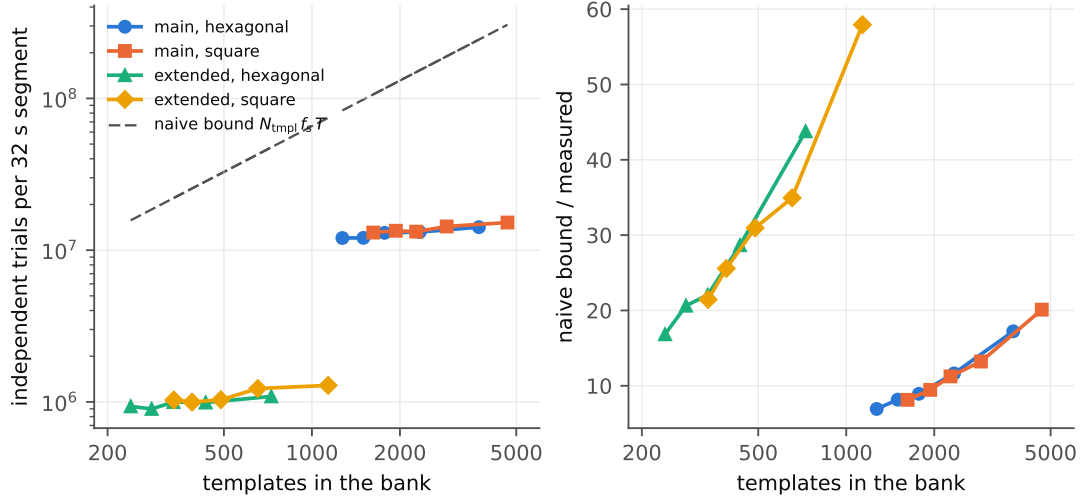


Figure 4: Effective independent trials measured from simulated noise, against the naive bound that counts every template and every sample. **How this could fail:** if a bank’s templates were independent the points would lie on the dashed line and the ratio would be 1; if trials grew in proportion to the template count each family would run parallel to it. Neither happens. An earlier version of this figure plotted ρ^* against MM; that was decoration, because ρ^* is a closed function of the template count, and it has been withdrawn.

bank triples, against 1.46 % naive, and the 2.48 % gain in $\langle M^3 \rangle$ then wins.

Two things must be said about that reversal. The pre-registered hypothesis — that the threshold rise and the finer bank balance somewhere between $\text{MM} = 0.97$ and 0.99 — is not supported either way, but the calibrated curves are flat to within about 2 % and are *not* monotonic in the extended region, so no optimum is resolved inside the grid. And the whole variation is smaller than the differences between the banks themselves. The honest reading is that within this model the minimal match barely moves the effective volume, and what weak preference exists is for finer banks. Where the optimum actually sits is then fixed by computational cost — the axis Keppel [8] optimises, and the one this work deliberately does not.

8 What was not done

- **No waveform check of the delivered banks.** Every covering number is the quadratic form. A sample of holes in an earlier bank was confirmed against real TaylorF2 matches, which found the quadratic form conservative rather than flattering, but none of the twenty banks reported here has been checked that way.
- **Cokelaer’s explicit connectors were never implemented.** Growth uses a proximity threshold as a stand-in, so the “lattice only” counts are not his lattice either.
- **The pre-registered placement sensitivities were not run** — a globally anchored metric, and discarding non-physical cells rather than projecting them.
- ν_{eff} **was calibrated at one segment length**, and extrapolated to a year as a rate. A half-segment check gives 0.432 against an expected 0.5.
- **The IMRPhenomD volume leg is an approximation**, composing a continuous-family fitting factor with the discretisation match and assuming the two are independent. The rigorous statement is the family-only bound, which is quoted beside it.

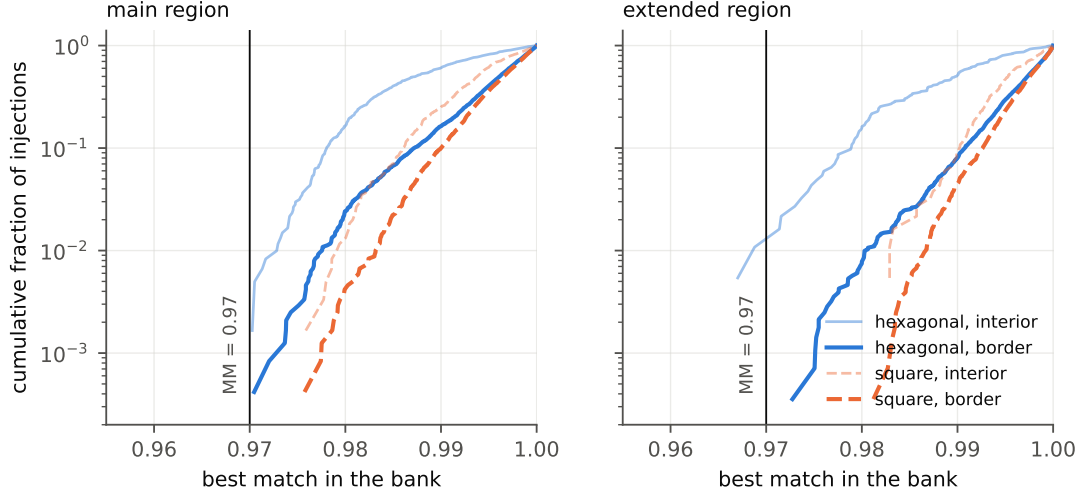


Figure 5: Cumulative distribution of the best match each injection finds, interior and border apart. **How this could fail:** a bank that did not cover would show a visible fraction of its curve left of the $MM = 0.97$ line. The extended hexagonal interior curve does cross it, by two injections in three thousand — and §5 explains why that number carries almost no information.

- **An independent cross-check of bank spacing by the SVD construction of [6]** was planned and cut.

9 Conclusion

The two questions this work set out to answer have asymmetric answers. How much SNR a mismatch costs is exact and settled: the loss is the mismatch, and above $M = 35 M_{\odot}$ the dominant loss is not the bank at all but the waveform model, at half the detection volume. How dense a bank must be has no answer in these terms, because within a plausible range the effective volume hardly depends on the density, and the direction of what little dependence there is turned out to be set by the trials model rather than by the bank. The practical content of the second answer is therefore negative, and worth stating plainly: the minimal match is not where the sensitivity of this search is won or lost.

Reproducing this. Every number above resolves through `\dataref` against `data/project_numbers.json`, whose sole writer is `scripts/compute_numbers.py`; every figure is produced by `scripts/make_figures.py` and by nothing else, from the tracked data in `data/. tests/test_acceptance.py` pins the reported values.

References

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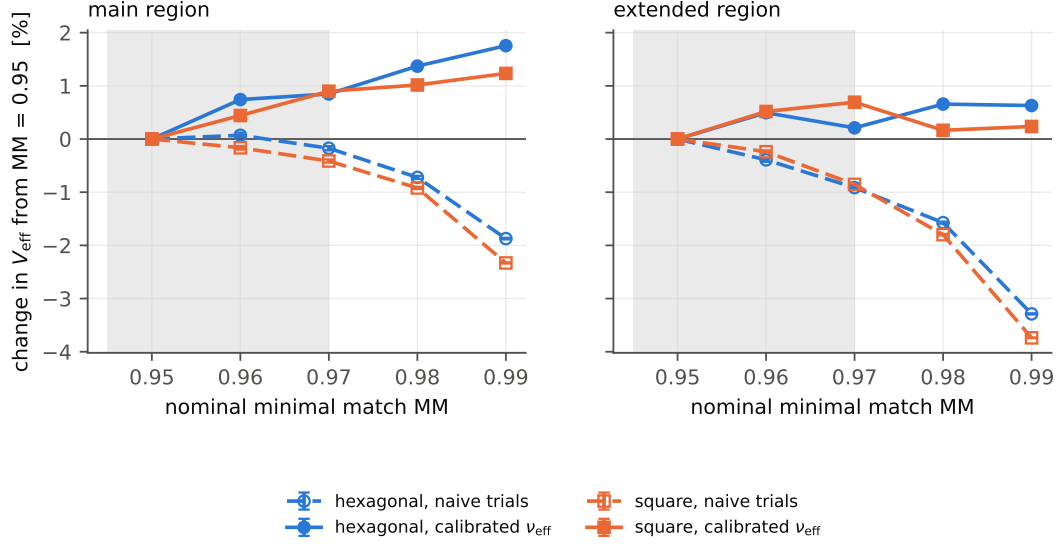


Figure 6: Effective volume under both trials models. **How this could fail:** if the two models agreed, the dashed and solid curves would coincide and the choice would not matter.

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